Financial Transactions Guard

Fraud Detection (CC4036)

Project 2026/2027

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Description

Financial institutions process millions of transactions daily, including payments, transfers, and currency exchanges involving different accounts, banks, and geographical locations. As financial systems become increasingly interconnected, detecting fraudulent activities and money laundering becomes more challenging. Fraudsters often exploit complex transaction patterns to conceal illicit activities, making automated detection essential for financial security.

Using a synthetic dataset that mimics real-world financial transactions, the goal of this project is to build, justify, and evaluate a machine learning–based solution for detecting suspicious transactions

FinTX-Guard Dataset

The FinTX-Guard (Financial Transactions Guard) Dataset contains simulated financial transactions designed to support the development and evaluation of machine learning methods for detecting suspicious financial activity.

Each instance (row) represents a single financial transaction described by features such as the transaction amount, payment type, sender and receiver accounts, bank locations, currencies, and transaction date and time.

The dataset includes a binary target variable, isFraud, indicating whether a transaction is associated with suspicious financial activity (isFraud = 1) or is legitimate (isFraud = 0).

The description of all the features that compose the dataset is presented in Table 1.

Tasks

Using the **FinTx-Guard_train.csv** data set, you have a set of main tasks to accomplish as described next. Still, you are free to include other tasks to increase the value of your assignment.

Task 1: Data Understanding and Preparation and Descriptive Modelling
This task involves summarizing and visualizing the data to provide valuable insights. Consider questions that could be interesting to check with the available data and provide answers using textual summaries or data visualization. Based on this analysis, you should also check if it is necessary to carry out any data clean-up and pre-processing steps. Additionally, you can apply clustering techniques if you find them helpful to provide valuable insights.

Table 1: Features included in FinSim-Mobile dataset.

Feature name	Description
transaction_id	Unique identifier assigned to each transaction.
time	Time at which the transaction was recorded (HH:MM.SS).
date	Date on which the transaction was recorded (YYYY-MM-DD).
sender_id	Anonymized identifier of the account initiating the transaction.
receiver_id	Anonymized identifier of the account receiving the transaction.
amount	Monetary amount associated with the transaction, expressed in the payment currency.
payment_currency	Currency in which the transaction was initiated.
received_currency	Currency in which the transaction was received.
sender_bank_location	Country or region where the sender's bank is located.
receiver_bank_location	Country or region where the receiver's bank is located.
payment_type	Type of financial transaction (e.g., bank transfer, credit card, cash withdrawal).
customer_session_duration	Duration associated with the customer's transaction session, in seconds.
account_activity_score	Numerical indicator representing the activity level associated with the sending account (higher values indicate greater activity).
transaction_channel	Channel used to initiate the transaction (e.g., Web, Mobile App, POS, Other).
device_recognized	Indicates whether the device associated with the transaction is recognized (1 = Yes, 0 = No).
device_risk_score	Numerical indicator representing the risk level associated with the device used for the transaction (higher values indicate greater risk).
verification_score	Numerical indicator representing the verification status or confidence associated with the transaction (higher values indicate greater confidence).
channel_flag	Binary indicator associated with the transaction channel or its processing context (1 = Yes, 0 = No).
isFraud	Binary indicator of suspicious financial activity (1 = suspicious, 0 = legitimate).

Task 2: Predictive Modelling
From the available data, you should define the data set used for the classification task at hand. Different algorithms should be considered, and the choice of the final model should be justified.

Task 3: Kaggle Competition
Additionally, you should submit your solution for the test data set that is available on the Kaggle Competition. Note that the goal is not to overfit the test set data to achieve a good position in the public leaderboard. Your rank position in the private leaderboard will be accounted for in the final grade.

Tools

You can use R and/or Python. You can find material for dynamic reporting in R with markdown if you choose to use R. You can use the Colab Research Notebooks or Jupyter Notebooks if you decide to use Python.

Deliverables

The project is mandatory, should be performed in groups of two students, and submitted on moodle with a compressed file containing the following items:

- **slides for presentation (PDF format)** focusing on the main issues of the project for a 10 minutes presentation; any additional information that cannot be presented in the 10 min slot can be included as annexes to the presentation;

- the source of **two ready-to-execute dynamic reports or notebooks** with all the code necessary to reproduce the results you present for i) Task 1; and ii) Task 2, including the solution submitted to Kaggle - Task 3, i.e. it should be possible to replicate your predictions; any complementary files needed to execute your report (e.g. data files, data objects) should be submitted as well.

Deadline

December 16th, 2026, 23:59

Presentation Guidelines

- problem definition (1 slide)
- data understanding: concise summary, focusing on the main findings (2 slides)
- data preparation: outline, focusing on the main operations (2 slides)
- predictive modelling with experimental results (2 slides)
- conclusions, limitations and future work (1 slide)
- annexes

TOTAL: max 40 slides

Grading

The grading of the practical assignment is distributed as follows:

- Task 1 (25%)
- Task 2 (25%)
- Task 3 (10%)
- Originality (15%)
- Presentation (25%) - **mandatory**